Soumiya_Ayyappan_W6_D2

Code and its output:

Step 1:

Code:

spark_basics.py | create_dataframe.py | employee_analytics.py | codecod.py

day1 > codecod.py > ...

```python
from pyspark.sql import SparkSession

spark=SparkSession.builder.appName("CustomerOrdersLab").getOrCreate()

from pyspark.sql.functions import (col,to_date,sum,count,year,month,dayofmonth)
from pyspark.sql.types import (StructType,StructField,StringType,IntegerType)

dataschema=StructType(
    [
        StructField("order_id",IntegerType(),True),
        StructField("order_date",StringType(),True),
        StructField("customer_id",StringType(),True),
        StructField("customer_name",StringType(),True),
        StructField("product_category",StringType(),True),
        StructField("product_name",StringType(),True),
        StructField("brand",StringType(),True),
        StructField("unit_price",IntegerType(),True),
        StructField("quantity",IntegerType(),True),
        StructField("payment_mode",StringType(),True),
        StructField("city",StringType(),True)
    ]
)
df=spark.read.option("header","true").schema(dataschema).csv("/home/kasm-user/pyspark_training/day1/data/coddataset.csv")
df.printSchema()
df.show()

```

Output:

```
root
 |-- order_id: integer (nullable = true)
 |-- order_date: string (nullable = true)
 |-- customer_id: string (nullable = true)
 |-- customer_name: string (nullable = true)
 |-- product_category: string (nullable = true)
 |-- product_name: string (nullable = true)
 |-- brand: string (nullable = true)
 |-- unit_price: integer (nullable = true)
 |-- quantity: integer (nullable = true)
 |-- payment_mode: string (nullable = true)
 |-- city: string (nullable = true)

+--------+----------+-----------+-------------+----------------+----------------+----------+----------+--------+------------+-
--------+
|order_id|order_date|customer_id|customer_name|product_category|     product_name|     brand|unit_price|quantity|payment_mode|
    city|
+--------+----------+-----------+-------------+----------------+----------------+----------+----------+--------+------------+-
--------+
|    1001|01/05/2025|       C001| Rahul Sharma|     Electronics|   Wireless Mouse|  Logitech|      1200|       2|         UPI|
  Mumbai|
|    1002|01/08/2025|       C002|  Priya Singh|        Clothing|            Jeans|     Levis|      2500|       1| Credit Card|
   Delhi|
|    1003|01/12/2025|       C003|   Amit Kumar|     Electronics|         Keyboard|        HP|      1800|       2|  Debit Card|B
angalore|
|    1004|01/15/2025|       C004|   Neha Patel|         Grocery|         Rice Bag|India Gate|       900|       3|         UPI|H
```

```
+--------+----------+-----------+-------------+---------------+----------------+----------+----------+--------+------------+-
--------+
|order_id|order_date|customer_id|customer_name|product_category|     product_name|     brand|unit_price|quantity|payment_mode|
    city|
+--------+----------+-----------+-------------+---------------+----------------+----------+----------+--------+------------+-
--------+
|    1001|01/05/2025|       C001| Rahul Sharma|    Electronics|   Wireless Mouse|  Logitech|      1200|       2|         UPI|
  Mumbai|
|    1002|01/08/2025|       C002|  Priya Singh|       Clothing|            Jeans|     Levis|      2500|       1| Credit Card|
   Delhi|
|    1003|01/12/2025|       C003|   Amit Kumar|    Electronics|         Keyboard|        HP|      1800|       2|  Debit Card|B
angalore|
|    1004|01/15/2025|       C004|   Neha Patel|        Grocery|         Rice Bag|India Gate|       900|       3|         UPI|H
yderabad|
|    1005|01/20/2025|       C005|  Arjun Mehta|       Clothing|          T-Shirt|      Puma|      1200|       2|        Cash|
    Pune|
|    1006|02/03/2025|       C006|    Sneha Rao|    Electronics|       Headphones|      Sony|      3500|       1| Credit Card|
  Mumbai|
|    1007|02/07/2025|       C007|  Rohan Verma|        Grocery|      Cooking Oil|   Fortune|       750|       4|         UPI|
   Delhi|
|    1008|02/11/2025|       C008|  Anjali Shah|       Clothing|         Sneakers|      Nike|      4500|       1|  Debit Card|B
angalore|
|    1009|02/18/2025|       C009| Vikram Joshi|    Electronics|      Smart Watch|      Boat|      2800|       2|         UPI|
    Pune|
|    1010|02/25/2025|       C010|   Kavya Nair|        Grocery|           Coffee|   Nescafe|       500|       5|        Cash|H
yderabad|
```

Step 2:

Code:

```
df1=df.withColumn("order_date",to_date(col("order_date"),"yyyy-MM-dd"))
df2=df1.withColumn("total_amount",col("unit_price")*col("quantity"))
df3=df2.withColumnRenamed("unit_price","price")
df4=df3.withColumn("order_year",year(col("order_date")))
df5=df4.withColumn("order_month",month(col("order_date")))
df6=df5.withColumn("order_day",dayofmonth(col("order_date")))

catgeory_sales=df6.groupBy("product_category").agg(sum("total_amount").alias("total_sales"))
catgeory_sales.show()

catgeory_sales.orderBy(col("total_sales").desc()).show()
```

Output:

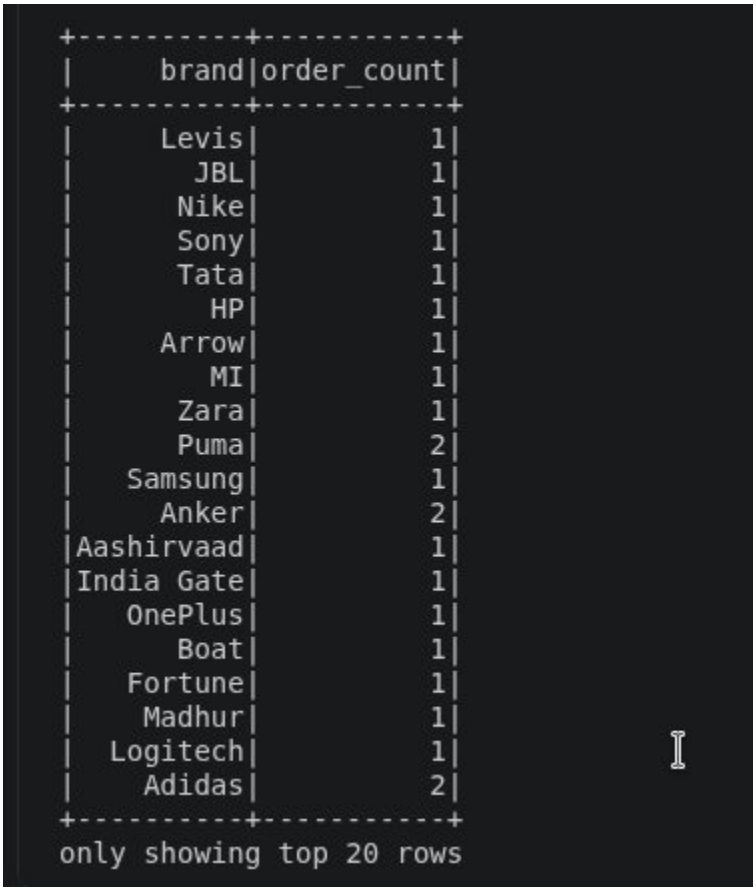
```
only showing top 20 rows
+----------------+-----------+
|product_category|total_sales|
+----------------+-----------+
|         Grocery|      17250|
|     Electronics|      46700|
|        Clothing|      26800|
+----------------+-----------+

+----------------+-----------+
|product_category|total_sales|
+----------------+-----------+
|     Electronics|      46700|
|        Clothing|      26800|
|         Grocery|      17250|
+----------------+-----------+
```

Step 3:

Code:

```
brand_orders=df6.groupBy("brand").agg(count("order_id").alias("order_count"))
brand_orders.show()
```



```
+----------+-----------+
|     brand|order_count|
+----------+-----------+
|     Levis|          1|
|       JBL|          1|
|      Nike|          1|
|      Sony|          1|
|      Tata|          1|
|        HP|          1|
|     Arrow|          1|
|        MI|          1|
|      Zara|          1|
|      Puma|          2|
|   Samsung|          1|
|     Anker|          2|
|Aashirvaad|          1|
|India Gate|          1|
|   OnePlus|          1|
|      Boat|          1|
|   Fortune|          1|
|    Madhur|          1|
|  Logitech|          1|
|    Adidas|          2|
+----------+-----------+
only showing top 20 rows
```

Step 4:

Code:

```
city_sales=df6.groupBy("city").agg(count("total_amount").alias("city_sales"))
city_sales.show()

city_sales.orderBy(col("city_sales").desc()).show()

payment_sales=df6.groupBy("payment_mode").agg(count("total_amount").alias("payment_sales"))
payment_sales.show()

monthly_orders=df6.groupBy("order_month").agg(count("order_id").alias("total_orders"))
monthly_orders.show()

final_df=df6.select(
    "order_id","order_date","customer_name","product_category","product_name",
    "brand","price","qunatity","total_amount","payment_mode","city"
)
final_df.show(truncate=False)
```

Output:

```
only showing top 20 rows
+---------+----------+
|     city|city_sales|
+---------+----------+
|Bangalore|         5|
|   Mumbai|         5|
|     Pune|         5|
|    Delhi|         5|
|Hyderabad|         5|
+---------+----------+

+---------+----------+
|     city|city_sales|
+---------+----------+
|Bangalore|         5|
|   Mumbai|         5|
|     Pune|         5|
|    Delhi|         5|
|Hyderabad|         5|
+---------+----------+
```

payment_mode	payment_sales
Credit Card	5
Cash	5
Debit Card	5
UPI	10

order_month	total_orders
1	5
3	5
5	5
4	5
2	5

order_id	order_date	customer_name	product_category	product_name	brand	price	quantity	total_amount	payment_mode	city
1001	2025-01-05	Rahul Sharma	Electronics	Wireless Mouse	Logitech	1200	2	2400	UPI	Mumbai
1002	2025-01-08	Priya Singh	Clothing	Jeans	Levis	2500	1	2500	Credit Card	Delhi
1003	2025-01-12	Amit Kumar	Electronics	Keyboard	HP	1800	2	3600	Debit Card	Bangalore
1004	2025-01-15	Neha Patel	Grocery	Rice Bag	India Gate	900	3	2700	UPI	Hyderabad
1005	2025-01-20	Arjun Mehta	Clothing	T-Shirt	Puma	1200	2	2400	Cash	Pune
1006	2025-02-03	Sneha Rao	Electronics	Headphones	Sony	3500	1	3500	Credit Card	Mumbai
1007	2025-02-07	Rohan Verma	Grocery	Cooking Oil	Fortune	750	4	3000	UPI	Delhi
1008	2025-02-11	Anjali Shah	Clothing	Sneakers	Nike	4500	1	4500	Debit Card	Bangalore
1009	2025-02-18	Vikram Joshi	Electronics	Smart Watch	Boat	2800	2	5600	UPI	Pune
1010	2025-02-25	Kavya Nair	Grocery	Coffee	Nescafe	500	5	2500	Cash	Hyderabad
1011	2025-03-02	Rahul Sharma	Electronics	USB Cable	Anker	800	3	2400	UPI	Mumbai
1012	2025-03-06	Priya Singh	Clothing	Jacket	Adidas	3200	1	3200	Credit Card	Delhi
1013	2025-03-10	Amit Kumar	Grocery	Tea	Tata	450	6	2700	Debit Card	Bangalore
1014	2025-03-15	Neha Patel	Electronics	Power Bank	MI	1500	2	3000	UPI	Hyderabad
1015	2025-03-22	Arjun Mehta	Clothing	Shirt	Arrow	2200	2	4400	Cash	Pune
1016	2025-04-04	Sneha Rao	Electronics	Bluetooth Speaker	JBL	4200	1	4200	Credit Card	Mumbai
1017	2025-04-09	Rohan Verma	Grocery	Flour	Aashirvaad	650	3	1950	UPI	Delhi
1018	2025-04-14	Anjali Shah	Clothing	Track Pants	Puma	1800	2	3600	Debit Card	Bangalore
1019	2025-04-20	Vikram Joshi	Electronics	Tablet	Samsung	18000	1	18000	UPI	Pune
1020	2025-04-28	Kavya Nair	Grocery	Biscuits	Parle	300	8	2400	Cash	Hyderabad

only showing top 20 rows